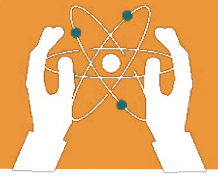


UNIT 8

UNDERSTANDING SIMPLE MACHINES

Learning objectives:

- Understand the functions and applications of simple machines like wheels, axles, pulleys, and gears.
- Differentiate between fixed, movable, and compound pulleys and their uses.
- Explain how gears increase speed or change force direction in mechanical systems.



UNDERSTANDING SIMPLE MACHINES

Have you ever used a see-saw, opened a door, or pulled water from a well? If yes, then you have used a simple machine! Simple machines make work easier by reducing the effort needed to move an object.

This chapter explores simple machines; fundamental tools that help us perform tasks more efficiently by altering forces. The focus will be on wheels and axles, pulleys, and gears

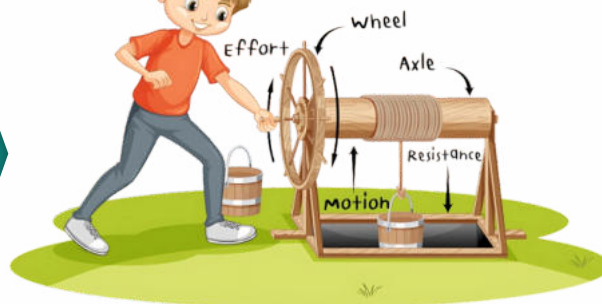
Types of Simple Machines

Wheel and Axle

A wheel and axle is a simple machine that consists of:

- A **wheel** (a circular object)
- An **axle** (a rod passing through the center of the wheel)

The wheel turns around the axle, making movement easier by reducing friction and increasing force.

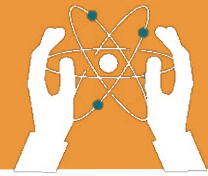


Wheel and Axle

• DID YOU KNOW?

Did you know that the wheel and axle were invented more than 5,000 years ago, revolutionizing transportation and machinery?





Applications:

Cars, Bicycles, and Motorcycles: The wheels help in smooth movement with less effort.

Trains and Buses: Use wheel and axle systems for easy rolling on tracks or roads.

Airplanes: Wheels help in takeoff and landing.

Cranes & Pulleys: Use large wheels and axles to lift heavy loads.

Door Knobs & Handles: The round knob (wheel) makes it easier to turn the latch (axle).

Pulleys

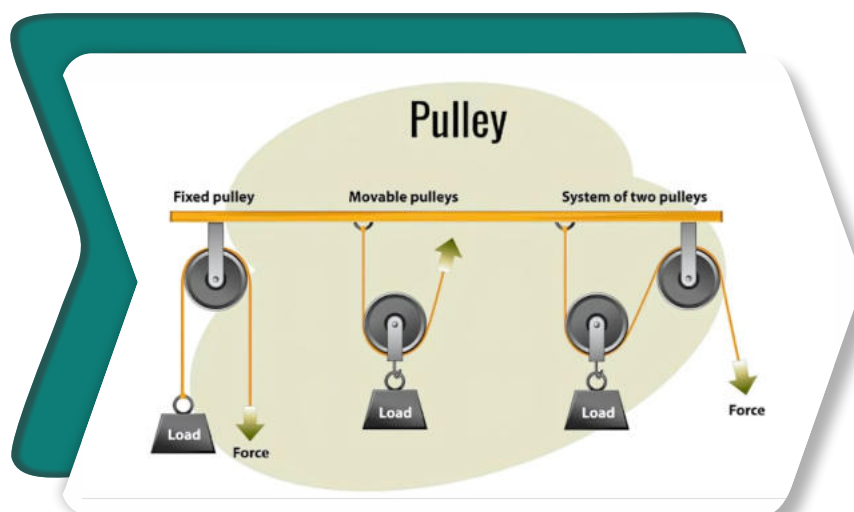
Pulley consist of a wheel on an axle with a groove around which a rope or cable can run to lift or move loads.

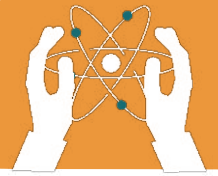
Types:

1. Fixed Pulley: It changes the direction of the force applied, making it easier to lift weights vertically.

2. Movable Pulley: It offers a mechanical advantage by allowing the effort force to be less than the load.

3. Compound Pulley: It combines fixed and movable pulleys to further reduce the effort needed.





Applications:

- **Wells:** A simple pulley helps in drawing water from wells.
- **Cable Cars & Ski Lifts:** Use pulley systems to transport people up slopes.
- **Ship Sails:** Pulleys help sailors adjust sails with less effort.
- **Flagpoles:** A pulley is used to raise and lower flags.
- **Zip Lines:** Pulleys help people slide down a cable for fun.

Activity: Create a pulley system using buckets and ropes to understand how different configurations (fixed, movable, and compound) affect the lifting force.

Gears

Gears are toothed wheels that fit together and rotate to transfer motion and force. They are a type of simple machine used to change speed, direction, or force in machines.

Functions:

Increase Speed: By connecting a larger gear to a smaller one, the smaller gear turns faster, increasing the speed of rotation.

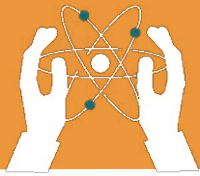
Change Direction: Gears can also change the direction of force, useful in mechanical clocks and automotive transmissions.



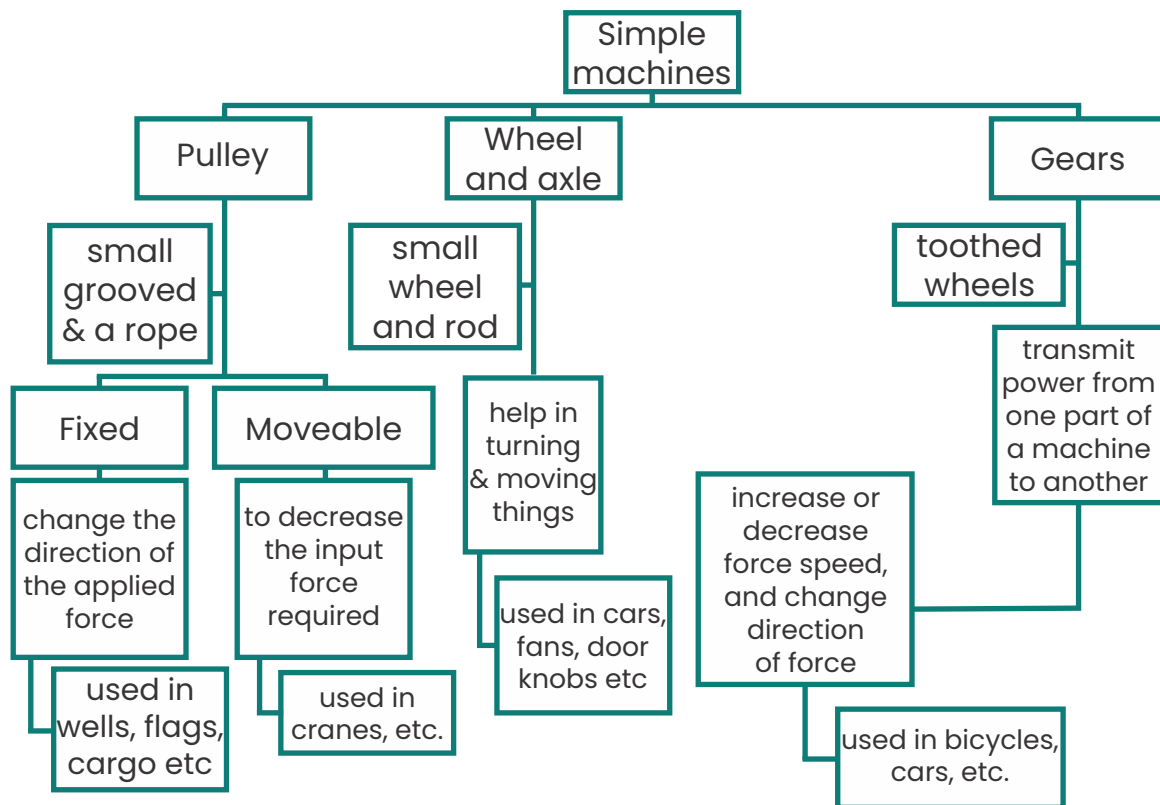
• DID YOU KNOW?

Pulleys have been used since ancient times, originating in Mesopotamia for lifting water and other heavy loads.





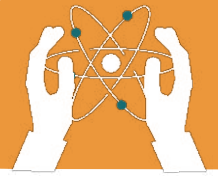
SUMMARY



• DID YOU KNOW?



- The wheel was one of the greatest inventions!
- Pulley systems can make you feel light!
- From scissors to bicycle gears, simple machines make life easier.
- Human hand behave like a simple machine.



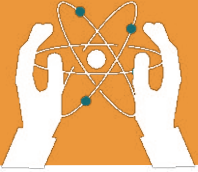
Answer the Following Questions:

Short-Answer Questions

- Q1. What is the primary function of a wheel and axle?
- Q2. Define pulley. Name the types of pulley.
- Q3. What is the purpose of a compound pulley?
- Q4. Define gears. How do gears increase speed?
- Q5. List two applications of pulleys.
- Q6. Write two differences between fixed and movable pulleys?

Long-Answer Questions

- Q1. Explain the differences between a fixed pulley, movable pulley, and compound pulley. Provide an example of each type.
- Q2. Describe the structure and function of a wheel and axle and give examples of where it is used.
- Q3. What are gears, and how do they function? Explain their role in increasing speed and changing direction with examples.
- Q4. Discuss the applications of pulleys in different fields and how they contribute to lift heavy objects.



UNDERSTANDING SIMPLE MACHINES

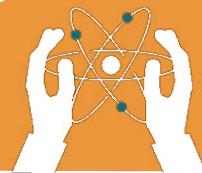
List the machines you may find in the bicycle.



Critical Thinking

Discuss how combining a wheel and axle with a pulley system could enhance the functionality of a mechanical device.

UNDERSTANDING SIMPLE MACHINES



Write functions of the given simple machines.

S.No.	Simple machines	Functions
1.	Pulley	
2.	Gear	
3.	Wheel and axle	

INQUIRY: Find out at least one example of each of pulley, wheel and axle, and gears in your home. Draw a picture of it, label its parts, and write what it is used for and how it is a simple machine.